

A GPU-based Constraint Programming Solver

JFPC 2026 (PAPER PRESENTED AT AAAI 2026)

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18th May 2026

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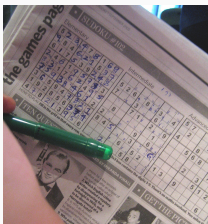


Constraint Programming

Constraint programming is a declarative approach to solve discrete constraint problems. In particular, it supports natively non-linear constraints.

$$x \in \{1, 2, 3\} \wedge y \in \{2, 3, 5\} \wedge x^2 - y = 5$$

A constraint solver can typically find a (best) solution satisfying the constraints.



What is the Problem?

Constraint solvers, and more largely combinatorial optimization, have not benefited yet from GPU architectures.

Why?

Support for GPU accelerate 2 251



Beneo Van

à or-tools-discuss

Occasionally, I found that I can apply for

Do you have plans to support GPUs?

Thanks



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No, gpu are good on dense structures.

Sat is sparse by nature.

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For 50 years+, constraint solvers have been primarily designed for CPU architectures.

Have we really tried on GPU architectures??

State of the Art: Combinatorial Optimization on GPU

Do we have an **exact** and **general-purpose** constraint solver **running on GPU**?

- **Incomplete**, general-purpose, full GPU: Often population-based algorithms¹.
- Complete, **not general**, full GPU: Specific algorithms²
- Complete, general-purpose, **hybrid CPU/GPU**:
 - offloading to GPU specialized filtering procedures^{3,4}.
 - **cuOpt**: new MILP solver—relaxation on GPU, search on CPU⁵.

¹A. Arbelaez and P. Codognet, *A GPU Implementation of Parallel Constraint-Based Local Search*, PDP, 2014.

²Jan Gmys. Exactly Solving Hard Permutation Flowshop Scheduling Problems on Peta-Scale GPU-Accelerated Supercomputers. *INFORMS Journal on Computing*, 2022.

³F. Campeotto et al., *Exploring the use of GPUs in constraint solving*, PADL, 2014

⁴F. Tardivo et al., *Constraint propagation on GPU: A case study for the AllDifferent constraint*, *Journal of Logic and Computation*, 2023.

⁵Using primal-dual linear programming (PDLP).

Turbo: a general and exact constraint solver fully executing on GPU (propagation + search).

Main Characteristics

- **Simple:** interval-based constraint solving + backtracking search (no global constraints, learning, restart, event-based propagation, ...).
- **Efficient?** Almost on-par with Choco on the quality of found objective bounds within 20mins. Can beat OR-Tools on some instances.



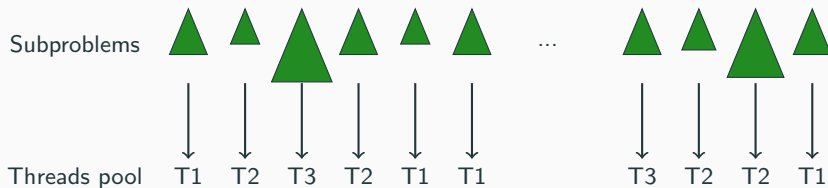
<https://github.com/ptal/turbo>

CPU-based Parallel Constraint Solvers

And Why The Same Techniques Do Not Work on GPU

On CPU: Embarrassingly Parallel Search (EPS)⁶

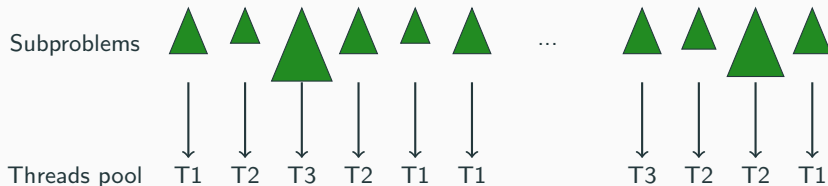
Divide the problem into many subproblems beforehand (e.g. $N \times 30$ with N the number of threads).



⁶A. Malapert et al., 'Embarrassingly Parallel Search in Constraint Programming', JAIR, 2016

On CPU: Embarrassingly Parallel Search (EPS)

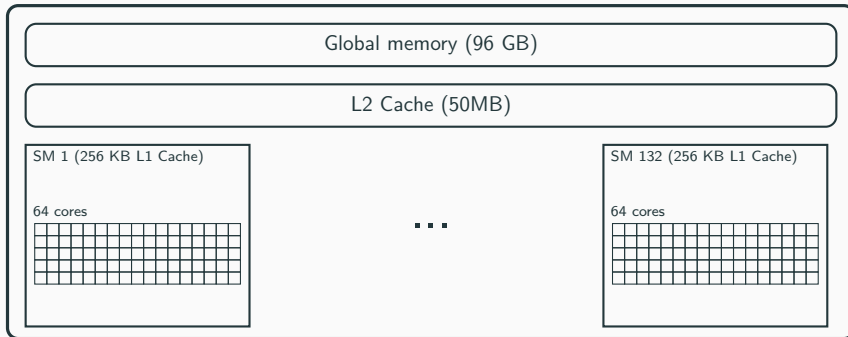
Divide the problem into many subproblems beforehand (e.g. $N \times 30$ with N the number of threads).



⇒ **Other approach:** portfolio approach (e.g., different search strategy on the *same problem*) as seen in Choco and OR-Tools.

Each thread works on its own copy of the problem.

On GPU: 1 Subproblem per Thread Takes too Much Memory



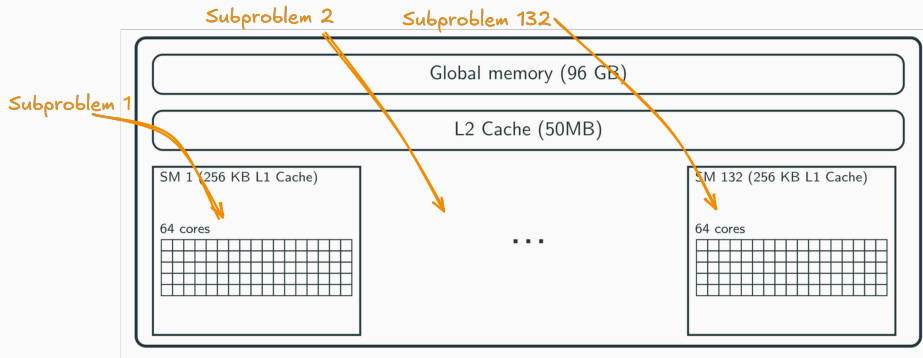
8448 cores grouped in 132 streaming multiprocessors (SM) of 64 cores each (H100).

⇒ **Oversubscribe** (to hide memory latency): 1024 threads per SM

135168 threads running in parallel!

For a 1MB constraint problem: 135GB of memory...

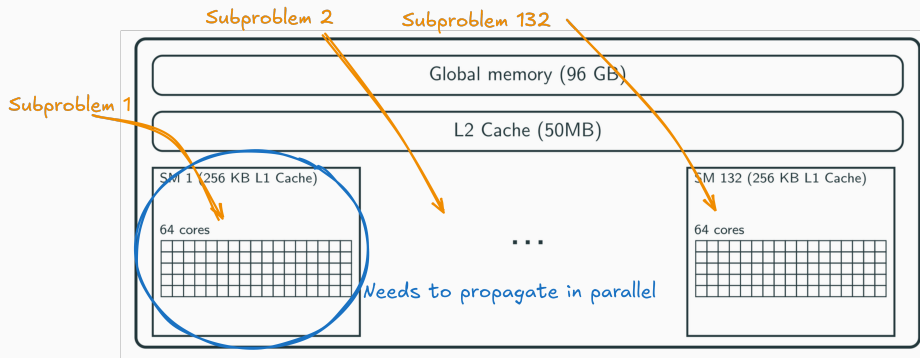
Search on GPU with EPS: one subproblem per SM⁶



To address load balancing issue, we create more subproblems than SMs (EPS).
(more details in the paper for the algorithm).

⁶More precisely, one subproblem per GPU block.

Propagation on GPU



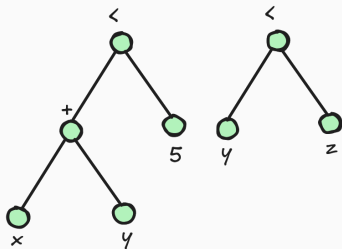
We proposed a correct model of lock-free parallel propagation⁷.

⁷P. Talbot et al., *A Variant of Concurrent Constraint Programming on GPU*, AAAI, 2022.

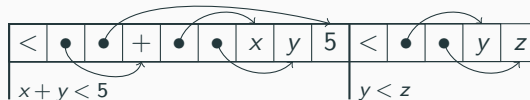
Representation of Constraints in AAI 2022

Constraints: $x + y < 5$ and $y < z$

Syntax tree



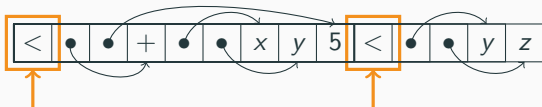
In memory



One Constraint per GPU Thread?

Constraint propagation is essentially traversing the syntax tree, but it is not efficient on GPU:

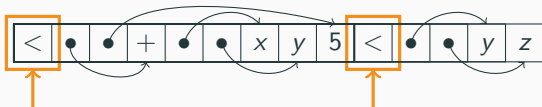
- **Non coalesced memory accesses:** 2 non-adjacent reads trigger 2 memory transactions.



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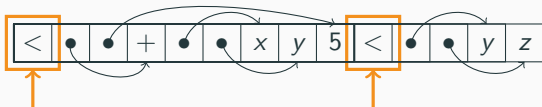


- **Load imbalance:** The first thread needs to do more work than the second.

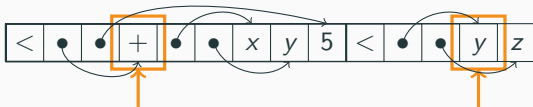
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- **Non coalesced memory accesses:** 2 non-adjacent reads trigger 2 memory transactions.



- **Load imbalance:** The first thread needs to do more work than the second.
- **Thread divergence:** The code path is different on different kind of nodes.



Our Solution: Ternary Constraint Network

Ternary Constraint Network

Each constraint takes a regular form $x = y \odot z$ with $x, y, z \in X$ (no constant) and $\odot \in \{+, /, *, \text{mod}, \text{min}, \text{max}, \leq, =\}$.

Example

Constraints: $x + y < 5 \rightsquigarrow t = x + y \wedge t < 5$
 $y < z \rightsquigarrow y = u + z \wedge u < 0$ (introduce auxiliary variables t and u)

In memory

+	t	x	y	+	y	u	z
$t = x + y$				$y = u + z$			

One Constraint per GPU Thread!

Ternary constraints are compactly stored and efficiently accessed:

- **Coalesced memory accesses:** Adjacent vectorized 16-byte loads.



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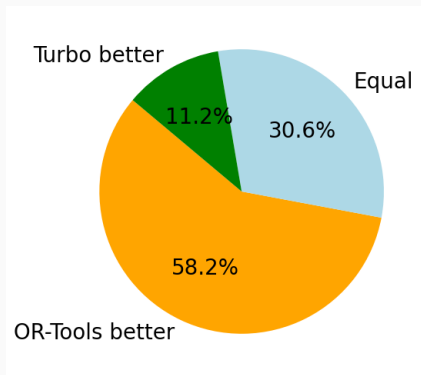
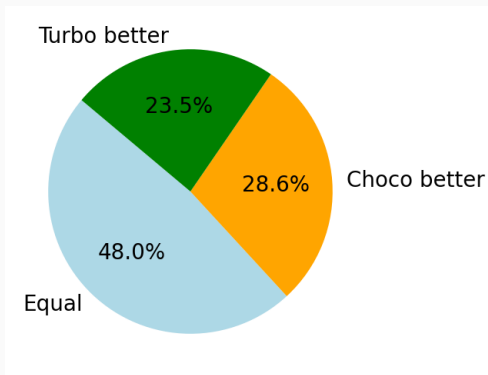
- **Uniform load balancing:** each thread performing the same work.
- **Thread divergence:** reduced by sorting constraints on $\odot$.

10x speed-up with ternary constraints.

Benchmark: 1-to-1 Comparison

On 98 instances of the MiniZinc 2024 competition.⁸

Comparison of the best objective values found. Timeout 20 minutes, GPU H100.



⁸1 instance unsat at root, 1 instance for which TCN is too large.

Turbo: General-purpose GPU constraint solver

- **Simple:** solving algorithms from 50 years ago.
⇒ no global constraints, nogoods learning, lazy clause generation, restart strategies, event-based propagation, trailing or recomputation-based state restoration and domain consistency.
- **Efficient:** Almost on-par with Choco (algorithmic optimization VS hardware optimization).
- Many possible optimizations to improve the efficiency, but need to be redesigned for GPU.



<https://github.com/ptal/turbo>

Next Steps of The Journey

- **New FNR Project:** *Neuro-Symbolic Reasoning for Embedded System Architectures Design*
LLM for constraint modelling ; Co-residence of CP solver and LLM on GPU.
- **GPU-based Continuous CSP:** Propagation over floating-point intervals.
Application to neural network verification.
- **Distributed CP:** Multi-GPUs using CRDTs⁹.
- **Parallel fixpoint loop improvement** (master students).



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Clément Poncelet



Manuel Combarro



Yi-Nung Tsao



Hakan Hasan



Wei Huang



Anisa Meta

⁹H. Hasan and P. Talbot, Towards Distributed Constraint Solving with CRDT, PaPoC 2026.

More Information...

Experimental Evaluation

On 98 instances of the MiniZinc 2024 competition.¹⁰

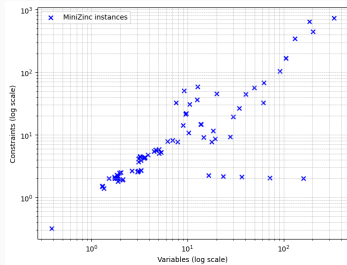
Timeout 20 minutes, CPU 64 cores, GPU H100.

solver	MiniZinc score	#Optimal
Or-Tools 9.9 (64 threads)	266.7	82
Choco 4.10.18 (64 threads)	190.8	44
Or-Tools 9.9 (fixed search)	119.9	37
Choco 4.10.18 (fixed search)	49.3	25
Turbo 1.2.8 (fixed search)	45.8	20

¹⁰1 instance unsat at root, 1 instance for which TCN is too large.

Drawback of Ternary Constraint Networks

Benchmark on the MiniZinc Challenge 2024 (96 instances)¹¹.

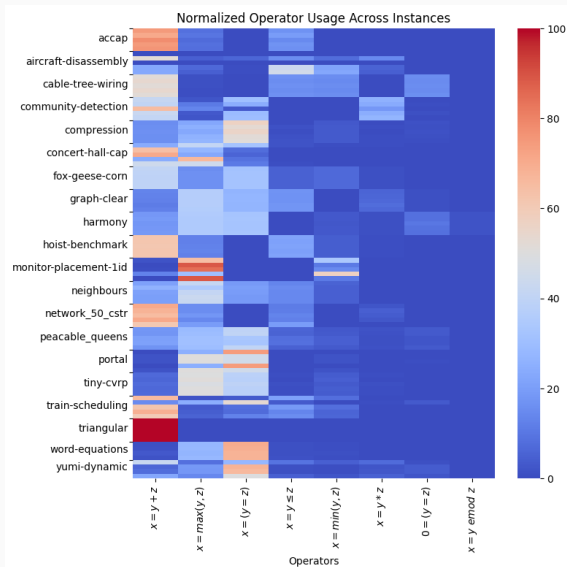


Increase in number of propagators and variables:

- The **median increase** of variables is 4.45x and propagators is 4.34x.
- The **maximum increase** of variables is 336x and propagators is 731x.

¹¹Instances not solved during preprocessing.

Divergence?



Lock-free Parallel Propagation

Example of Parallel Propagation¹²

Let's consider $\mathcal{I}[\![x \leq 4 \wedge x \leq 5]\!] = \mathcal{I}[\![x \leq 4]\!] \parallel \mathcal{I}[\![x \leq 5]\!]$

Memory:

$$x = [-\infty, \infty]$$

Propagators:

$$\begin{array}{ll} \boxed{x} \leftarrow [-\infty, 4] & (\mathcal{I}[\![x \leq 4]\!]) \\ \parallel \quad \boxed{x} \leftarrow [-\infty, 5] & (\mathcal{I}[\![x \leq 5]\!]) \end{array}$$

¹²P. Talbot et al., *A Variant of Concurrent Constraint Programming on GPU*, AAI, 2022.

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Memory:

$x = [-\infty, ?]$

Propagators:

$x \leftarrow [-\infty, 4] \quad (\mathcal{I}[\![x \leq 4]\!])$
 $\parallel \quad x \leftarrow [-\infty, 5] \quad (\mathcal{I}[\![x \leq 5]\!])$

Issue: nondeterminism? x can be equal to $[-\infty, 4]$ or $[-\infty, 5]$ depending on the order of execution.

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$\Rightarrow$ **Solution:** fixpoint + fair scheduling + strict updates (if $(v < x.\text{ub})$ { $x.\text{ub} = v$; }).

¹²P. Talbot et al., *A Variant of Concurrent Constraint Programming on GPU*, AAAI, 2022.

GPU Fixpoint Algorithm

```
__device__ void fixpoint(Store& d, Props* props, int n) {  
    __shared__ bool has_changed = true;  
    // Keep going until no variable domain is modified.  
    while(has_changed) {  
        __syncthreads(); has_changed = false; __syncthreads();  
    }
```


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    // Keep going until no variable domain is modified.  
    while(has_changed) {  
        __syncthreads(); has_changed = false; __syncthreads();  
        // Execute all propagators (similar to AC1)  
        for(int i = threadIdx.x; i < n; i += blockDim.x) {  
            has_changed |= props[i].propagate(d);  
        }  
        __syncthreads();  
    }  
}
```

GPU Challenges

- Coalesced memory accesses of the propagator representation `props[i]`.
- Avoiding divergence in `propagate`.